

Lesson Notes – Loops

Lesson Introduction

In Scratch, a **loop** is a block of code that **repeats an action multiple times**.

- Loops help make projects **efficient** because you don't have to write the same code again and again.
- Loops are used in **animations, games, and interactive stories** to repeat movements, sounds, or checks for events.

Learning Objectives

By the end of this lesson, learners will be able to:

1. Identify different types of loops in Scratch (**forever, repeat, repeat until**).
2. Use loops to **repeat actions** for sprites.
3. Combine loops with **motion, looks, and sound blocks**.
4. Understand how loops make a project **efficient and interactive**.
5. Apply loops to create **animations, games, or patterns**.

Key Vocabulary

- **Loop:** A set of instructions that repeats multiple times.
- **Forever:** Repeats an action endlessly until the project stops.
- **Repeat:** Repeats an action a specific number of times.
- **Repeat Until:** Repeats an action until a certain condition becomes true.
- **Iteration:** Each time the loop runs is called an iteration.

Step-by-Step Guided Activity

Step 1: Set Up

1. Open Scratch 3 and click **Create**.
2. Add a sprite (e.g., Cat) and a backdrop.

Step 2: Using a Repeat Loop

1. Make the sprite move 10 steps, 5 times:

```
when green flag clicked
repeat 5
  move 10 steps
  turn 15 degrees
end
```

The sprite will move and turn 5 times, creating a pattern.

Step 3: Using a Forever Loop

1. Make the sprite continuously bounce:

```
when green flag clicked
forever
  move 10 steps
  if on edge, bounce
end
```

The sprite keeps moving and bouncing off the edges of the stage indefinitely.

Step 4: Using Repeat Until Loop

1. Make the sprite move until it touches a color:

```
when green flag clicked
repeat until <touching color [red]?>
  move 10 steps
end
say "I reached red!" for 2 seconds
```

The sprite moves repeatedly until it touches the red color, then stops and says a message.

Step 5: Optional Enhancements

- Add **sounds** inside loops to create music or effects.
- Combine loops with **variables** to count scores or steps.
- Create **patterns** by combining loops with motion and looks.
- Add **conditional statements inside loops** to make the sprite react to obstacles or keys.

Discussion / Questions

1. What is a loop in Scratch?
2. What is the difference between a `repeat` loop and a `forever` loop?
3. How does a `repeat until` loop work?
4. How can loops make coding faster and easier?
5. Can you use loops to create animations or games?

Learning Outcome

By the end of this lesson, learners will be able to:

- Use **different loops** to repeat actions in Scratch projects.
- Combine loops with **motion, looks, sound, and conditionals**.
- Build **efficient, interactive games, animations, and patterns** using loops.